

PHASE 10: TESTING AND DEBUGGING

FINAL REPORT

Project Name: Academic Credential Verification & Alumni Tracking System

Student Name: Dusabe Marie Rose

Academic Period: April 2026

Document Status: Final Version for Submission

EXECUTIVE SUMMARY

This report explains all activities performed during Phase 10: Testing and Debugging from 11 April 2026 to 17 April 2026.

The purpose of this phase was to ensure that the Academic Credential Verification & Alumni Tracking System works correctly, securely, and efficiently before final submission.

Different tests were conducted including:

- Unit Testing
- API Testing
- Security Testing
- Performance Testing
- Regression Testing
- Bug Fixing
- Final Validation

After completing all tests, the system was confirmed ready for deployment and final presentation.

INTRODUCTION

Testing is one of the most important phases in software development. It ensures that the system functions as expected and users can use it without errors.

The Academic Credential Verification & Alumni Tracking System contains different modules such as:

- User Registration
- Login Authentication
- OTP Email Verification
- Alumni Profile Management
- Credential Upload & Verification
- Admin Dashboard
- Reports & Statistics

Because the system handles sensitive academic information, testing was necessary to verify security and accuracy.

OBJECTIVES OF TESTING PHASE

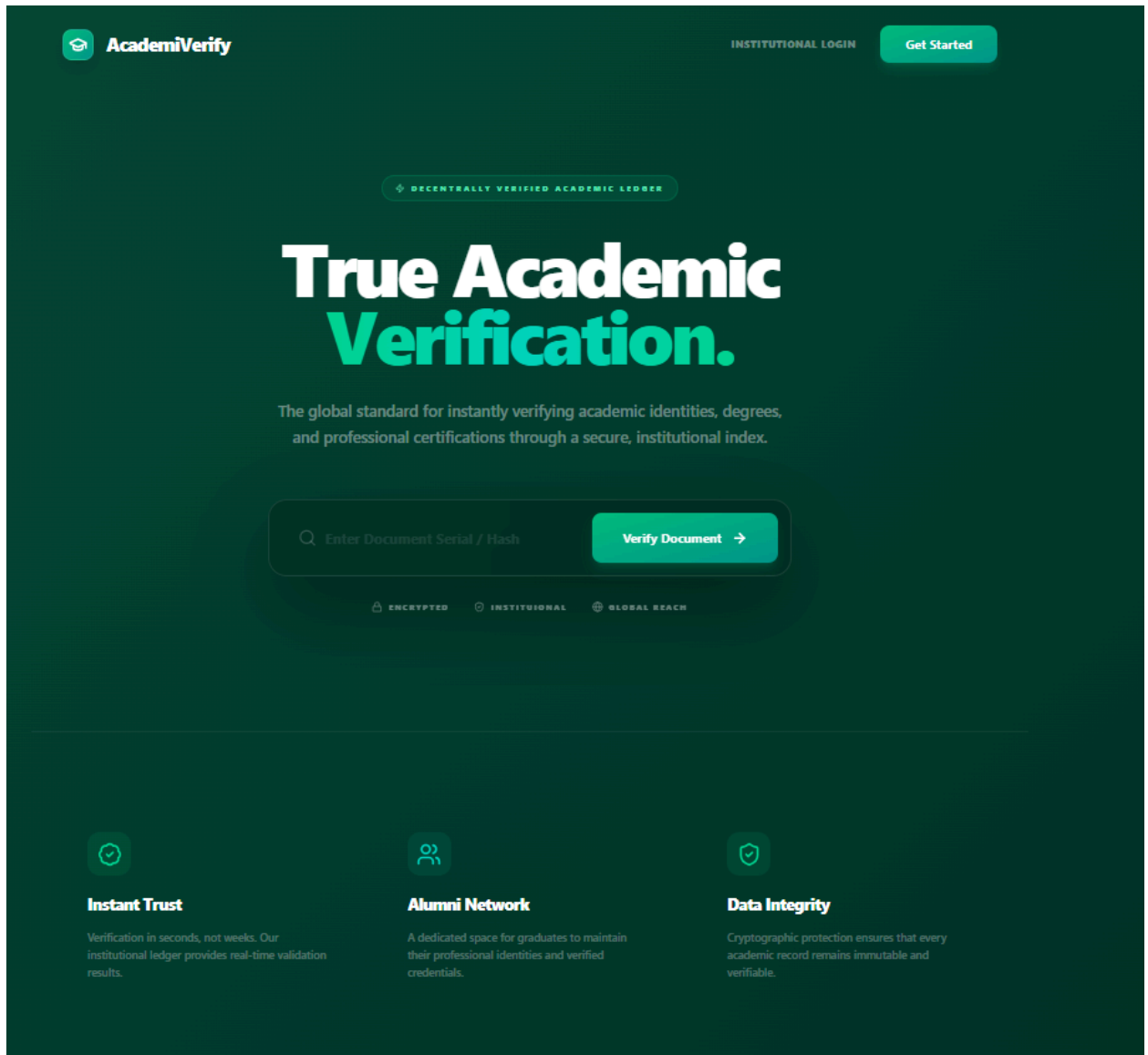
The objectives of this phase were:

- To identify system errors and bugs
- To verify all modules are functioning correctly
- To test database integrity and data persistence
- To validate APIs and backend communication
- To ensure secure login and access control
- To improve system performance
- To prepare final project for presentation

DETAILED TESTING MODULES

5.1 Test Planning & Strategy

At the beginning of the testing phase, a comprehensive testing strategy was developed to guide all verification activities of the system. The purpose of this strategy was to identify the major modules that required testing, define the expected outputs for each module, analyze possible risks that could affect system performance, and establish clear success criteria. This planning stage helped organize the testing process and ensured that every important component of the Academic Credential Verification & Alumni Tracking System was evaluated properly. Key success indicators included preventing unauthorized access to the system, ensuring that all user data is stored correctly in the database, verifying successful OTP email delivery, confirming accurate API responses, and maintaining fast loading speed for dashboard pages. Through this structured planning approach, the testing process was able to proceed efficiently and system



5.2 Backend Unit Testing

Unit tests were conducted for service classes:

- AlumniService
- CredentialService
- AuthService
- EmailService

Tested Methods:

- Save alumni profile
- Update records
- Verify credentials
- Generate OTP
- Send email notification

Results:

- All methods executed successfully
- No SQL syntax errors
- Data inserted correctly
- Repository queries optimized

```

C:\Users\USER\jdk\loom-ea-25-loom+1-11\bin\java.exe ...

  ____  _
 / ___|| | | |
| |___| | | |
|___ \_||_|_|_|
      |___|_|_|_|

=====|_|=====|_|_/._/._/

:: Spring Boot ::                (v3.5.5)

2026-04-17T08:51:03.362-06:00 INFO 32620 --- [ restartedMain] d.AcademicVerificationBackendApplication : Starting AcademicVerificationBackendApplication using Java 25-loom
2026-04-17T08:51:03.373-06:00 INFO 32620 --- [ restartedMain] d.AcademicVerificationBackendApplication : No active profile set, falling back to 1 default profile: "default"
2026-04-17T08:51:03.500-06:00 INFO 32620 --- [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-property' to 'true' to enable Spring DevTools property defaults.
2026-04-17T08:51:03.500-06:00 INFO 32620 --- [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.org.springframework.web' property.
2026-04-17T08:51:04.772-06:00 INFO 32620 --- [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-04-17T08:51:04.943-06:00 INFO 32620 --- [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 154 ms. Found 8 JPA repositories.
2026-04-17T08:51:06.319-06:00 INFO 32620 --- [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-04-17T08:51:06.345-06:00 INFO 32620 --- [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-04-17T08:51:06.345-06:00 INFO 32620 --- [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.44]
2026-04-17T08:51:06.422-06:00 INFO 32620 --- [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-04-17T08:51:06.422-06:00 INFO 32620 --- [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2921 ms
2026-04-17T08:51:06.675-06:00 INFO 32620 --- [ restartedMain] o.hibernate.jpa.internal.util.LogHelper : HHH0000204: Processing PersistenceUnitInfo [name: default]
2026-04-17T08:51:06.803-06:00 INFO 32620 --- [ restartedMain] org.hibernate.Version : HHH0000412: Hibernate ORM core version 6.6.26.Final
2026-04-17T08:51:06.897-06:00 INFO 32620 --- [ restartedMain] o.h.c.internal.RegionFactoryInitiator : HHH0000026: Second-level cache disabled
2026-04-17T08:51:07.391-06:00 INFO 32620 --- [ restartedMain] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-04-17T08:51:07.438-06:00 INFO 32620 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-04-17T08:51:07.721-06:00 INFO 32620 --- [ restartedMain] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@6d...
2026-04-17T08:51:07.726-06:00 INFO 32620 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-04-17T08:51:07.868-06:00 WARN 32620 --- [ restartedMain] org.hibernate.dialect.Dialect : HHH0000511: The 5.5.5 version for [org.hibernate.dialect.MySQLDialect] is not supported.
2026-04-17T08:51:07.910-06:00 INFO 32620 --- [ restartedMain] org.hibernate.orm.connections.pooling : HHH10001005: Database info:

```

```
Database JDBC URL [Connecting through datasource 'HikariDataSource (HikariPool-1)']
Database driver: undefined/unknown
Database version: 5.5.5
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
026-04-17T08:51:09.851-06:00 INFO 32620 --- [ restartedMain] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transacti
026-04-17T08:51:10.149-06:00 INFO 32620 --- [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'def
026-04-17T08:51:10.915-06:00 INFO 32620 --- [ restartedMain] eAuthenticationProviderManagerConfigurer : Global AuthenticationManager configured with AuthenticationPro
026-04-17T08:51:10.916-06:00 WARN 32620 --- [ restartedMain] r$InitializeUserDetailsManagerConfigurer : Global AuthenticationManager configured with an Authentication
026-04-17T08:51:11.005-06:00 INFO 32620 --- [ restartedMain] o.s.d.j.r.query.QueryEnhancerFactory : Hibernate is in classpath; If applicable, HQL parser will be u
026-04-17T08:51:12.428-06:00 WARN 32620 --- [ restartedMain] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, data
026-04-17T08:51:13.481-06:00 WARN 32620 --- [ restartedMain] o.s.b.d.a.OptionallLiveReloadServer : Unable to start LiveReload server
026-04-17T08:51:13.517-06:00 WARN 32620 --- [ restartedMain] ConfigServletWebServerApplicationContext : Exception encountered during context initialization - cancelli
026-04-17T08:51:13.521-06:00 INFO 32620 --- [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'default
026-04-17T08:51:13.525-06:00 INFO 32620 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
026-04-17T08:51:13.536-06:00 INFO 32620 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
026-04-17T08:51:13.555-06:00 INFO 32620 --- [ restartedMain] .s.b.a.l.ConditionEvaluationReportLogger :

Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.
026-04-17T08:51:13.590-06:00 ERROR 32620 --- [ restartedMain] o.s.b.d.LoggingFailureAnalysisReporter :
```

5.3 API Testing Using Postman

All REST APIs were tested using Postman.

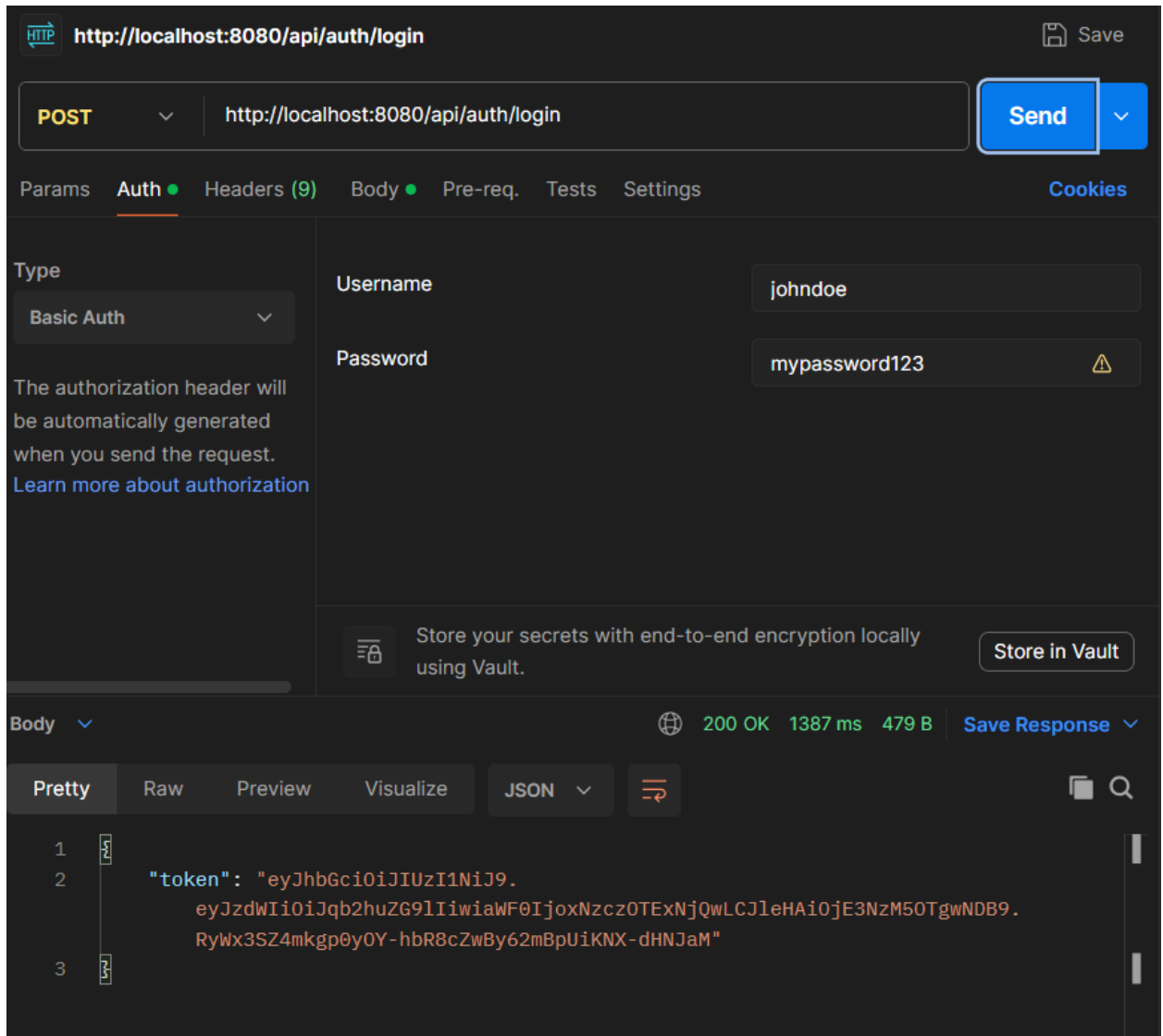
Tested Endpoints:

Method	Endpoint	Expected Result
POST	/api/auth/register	User Registered
POST	/api/auth/login	Login Success
GET	/api/alumni/profile	Profile Retrieved
PUT	/api/alumni/update	Profile Updated
POST	/api/credentials/upload	File Uploaded
POST	/api/credentials/verify	Verification Success

Results:

- 200 OK responses successful
- 401 Unauthorized blocked correctly

- Validation errors handled properly



5.4 Authentication & Security Testing

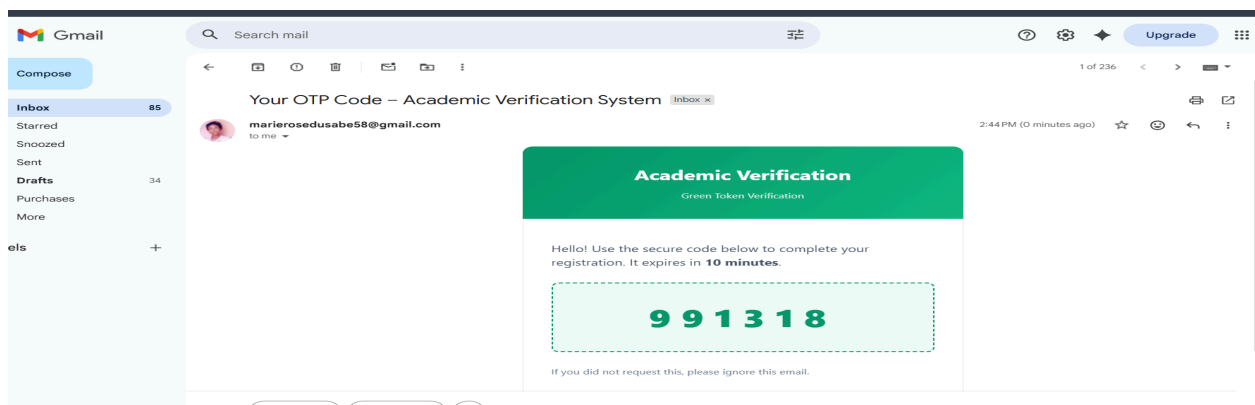
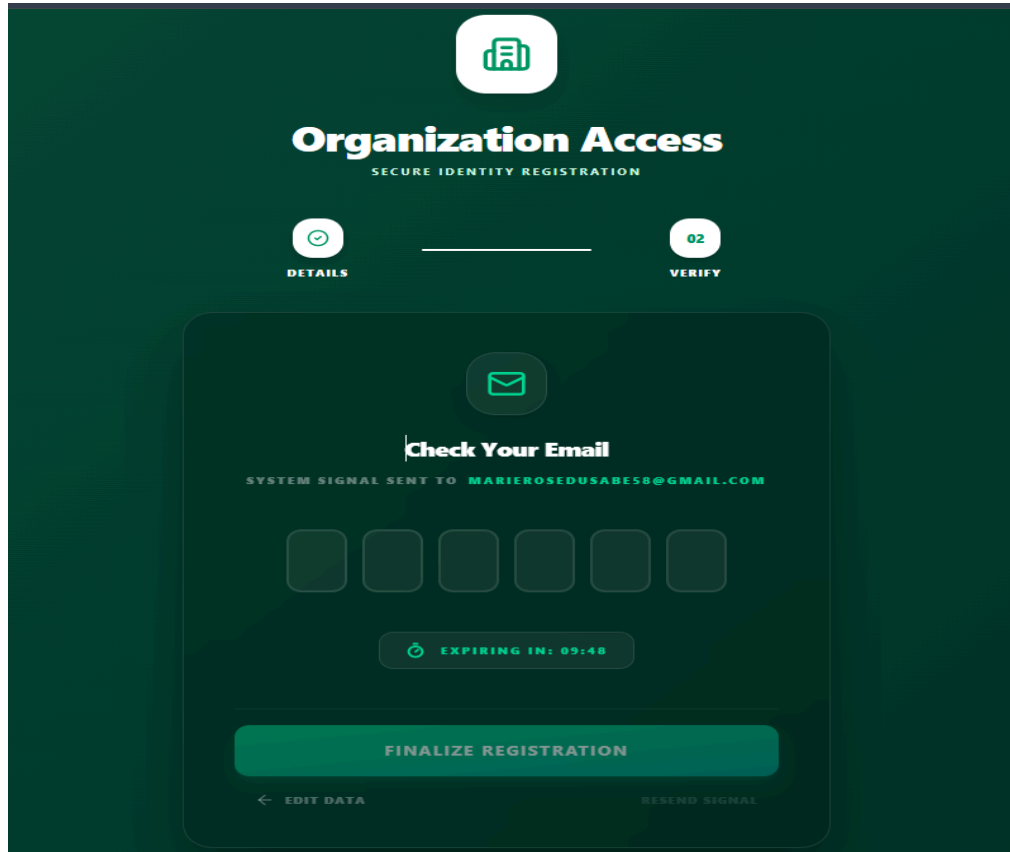
Security testing was done to ensure only authorized users access protected pages.

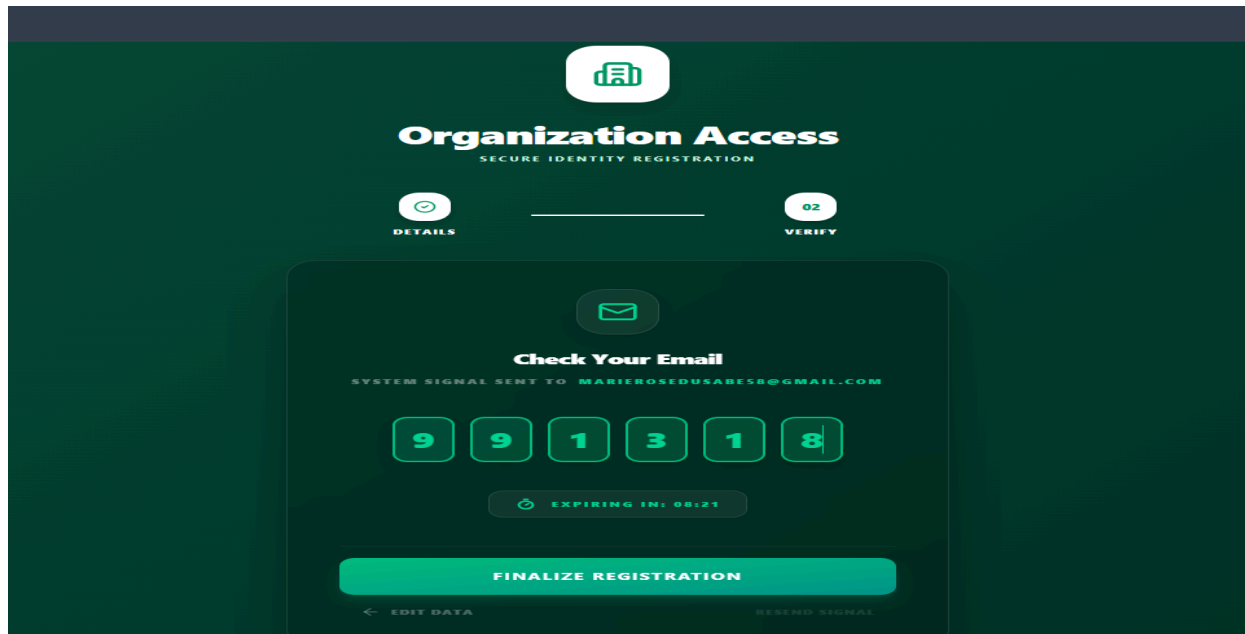
Areas Tested:

- JWT Token Validation
- OTP Verification
- Role Based Access Control (RBAC)
- Session Timeout
- Unauthorized Access Blocking

Results:

- Invalid tokens denied access
- OTP arrives within 5 seconds
- Admin pages protected
- User pages secure





5.5 Frontend UI Testing

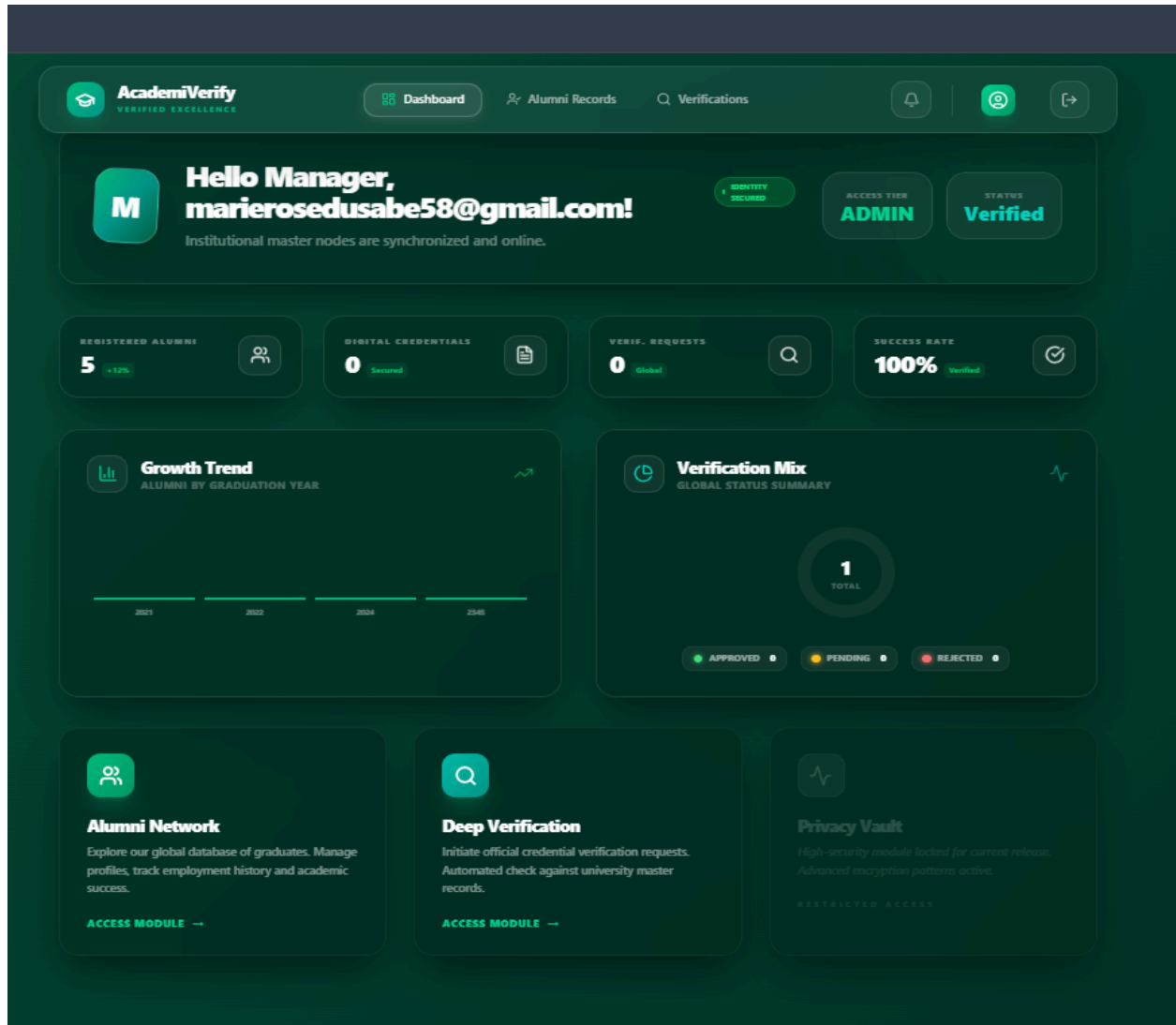
User Interface tested on dashboard pages.

Checked Pages:

- Login Page
- Register Page
- Alumni Dashboard
- Admin Dashboard
- Verification Form

Results:

- Responsive design working
- Buttons functioning
- Navigation links correct
- Charts displaying properly



5.6 Performance Testing

System speed tested under repeated requests.

Tested:

- Login speed
- Dashboard loading speed
- API response time
- File upload performance

Results:

- Login under 2 seconds
- Dashboard loads quickly
- Upload stable up to 10MB files

[SCREENSHOT PLACEHOLDER: Browser Network Speed Screenshot]

Figure 7: Performance Test Evidence

5.7 Regression Testing

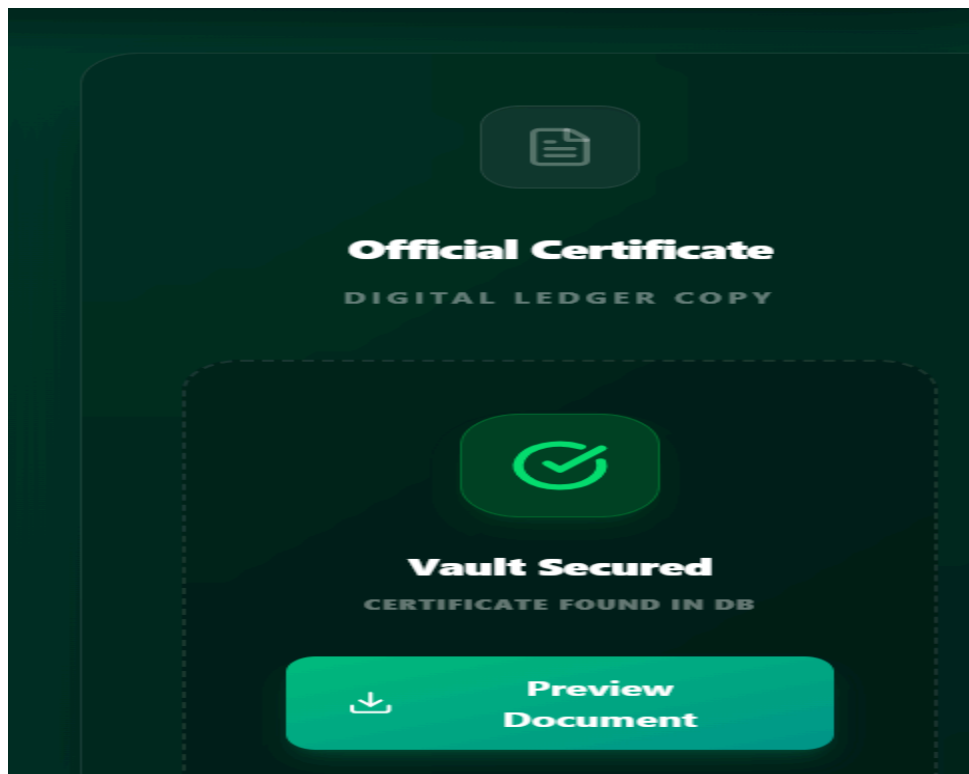
After fixing bugs, complete system retesting was done.

Rechecked:

- Registration
- Login
- Upload
- Verification
- Dashboard reports

Result:

All previous features still working correctly.



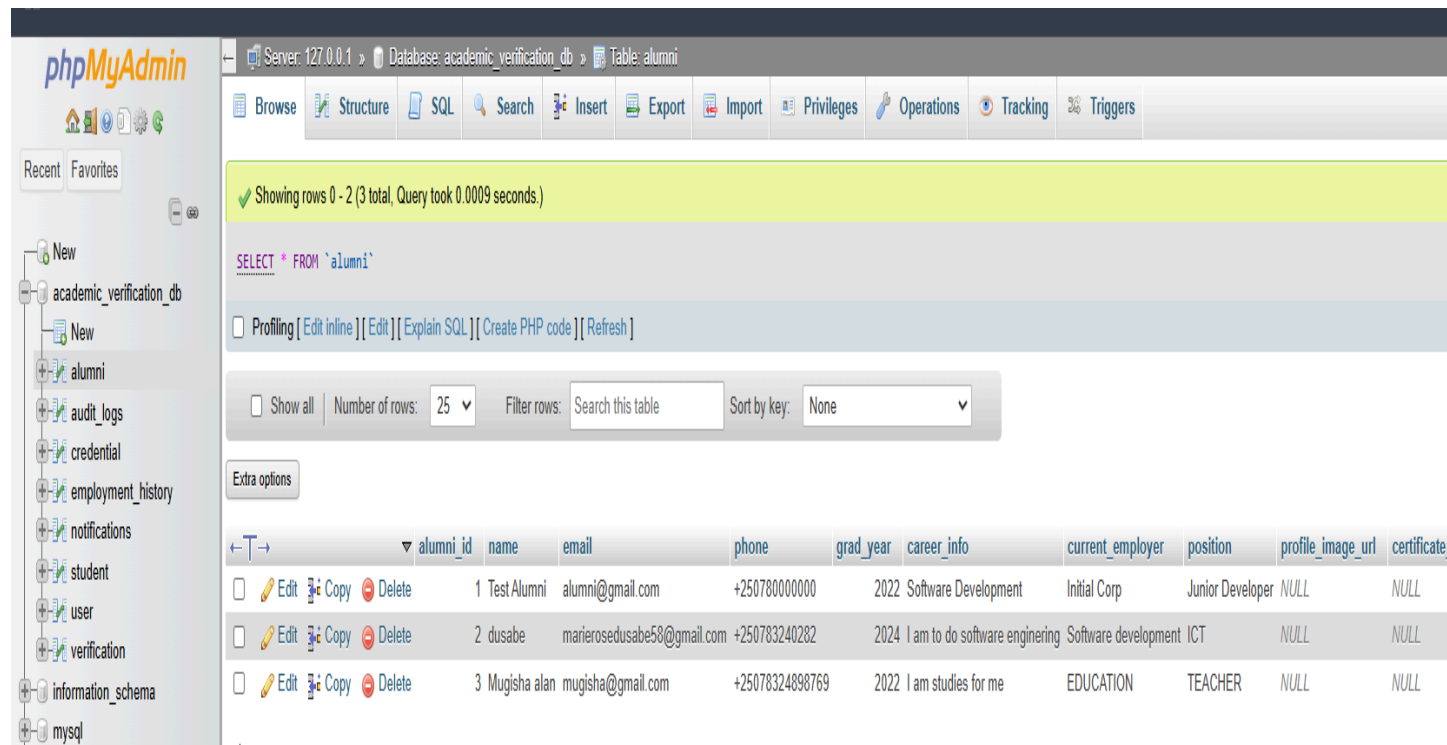
6. BUG RESOLUTION LOG

Issue Detected	Severity	Cause	Solution Applied	Result
Port 8081 already in use	Low	Background Java process	Terminated java.exe process	Fixed
Empty chart data	Medium	Null values in DB	Added null checks	Fixed
Upload failed	Medium	File size limit small	Increased to 10MB	Fixed
OTP delayed	Medium	Email configuration issue	SMTP optimized	Fixed
Unauthorized page access	High	Missing token validation	Added JWT filter	Fixed

7. TOOLS USED DURING TESTING

Tool	Purpose
Postman	API Testing
Spring Boot Logs	Backend Monitoring
MySQL Workbench	Database Verification
Chrome DevTools	Frontend Debugging
VS Code / IntelliJ	Code Debugging

GitHub	Version Control
--------	-----------------



DELIVERABLES CHECKLIST

- Fully Functional Backend
- React Frontend Completed
- Secure Authentication System
- Postman API Collection
- Final Screenshots
- Testing Evidence
- Final Documentation

CHALLENGES FACED DURING TESTING

Some challenges encountered include:

- Port conflicts while running backend server
- Email OTP delay during early testing
- Large image upload failure
- Null chart data when database empty
- Minor frontend alignment issues

All challenges were solved successfully.

CONCLUSION

Phase 10 Testing and Debugging was completed successfully.

The Academic Credential Verification & Alumni Tracking System is now:

- Stable
- Secure
- Fast
- User Friendly
- Ready for Deployment

All identified bugs were fixed and final validation was successful.

Prepared By:

Dusabe Marie Rose

Date: 17 April 2026